1. Library Management System

```
import java.util.Scanner;

class LibraryBook {
  int bookId;
  String bookTitle;
  String authorName;
  int copiesAvailable = 20;

  void readDetails() {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter Book ID: ");
    bookId = sc.nextInt();
    sc.nextLine();

    System.out.print("Enter Book Title: ");
    bookTitle = sc.nextLine();

    System.out.print("Enter Author Name: ");
    authorName = sc.nextLine();
  }

  void issueBook() {
    if (copiesAvailable > 0) {
      copiesAvailable--;
      System.out.println("Book Issued Successfully.");
    } else {
      System.out.println("No Copies Available.");
    }
  }

  void returnBook() {
    copiesAvailable++;
    System.out.println("Book Returned Successfully.");
  }
```

```
  void display() {
    System.out.println("\n----- Book Details -----");
    System.out.println("Book ID: " + bookId);
    System.out.println("Book Title: " + bookTitle);
    System.out.println("Author Name: " + authorName);
    System.out.println("Available Copies: " + copiesAvailable);
  }

  public static void main(String[] args) {
    LibraryBook b = new LibraryBook();
    b.readDetails();
    b.issueBook();
    b.returnBook();
    b.display();
  }
}
```

---

2. Employee Salary Management

```
import java.util.Scanner;

class Employee {
  int employeeId;
  String employeeName;
  String department;
  double basicSalary = 25000;

  double hra, da, grossSalary;

  void readDetails() {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter Employee ID: ");
    employeeId = sc.nextInt();
    sc.nextLine();

    System.out.print("Enter Employee Name: ");
```

```
        employeeName = sc.nextLine();

        System.out.print("Enter Department: ");
        department = sc.nextLine();
    }

    void calculateSalary() {
        hra = basicSalary * 0.20;
        da = basicSalary * 0.10;
        grossSalary = basicSalary + hra + da;
    }

    void display() {
        System.out.println("\n----- Employee Details -----");
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Employee Name: " + employeeName);
        System.out.println("Department: " + department);
        System.out.println("Basic Salary: ₹" + basicSalary);
        System.out.println("HRA: ₹" + hra);
        System.out.println("DA: ₹" + da);
        System.out.println("Gross Salary: ₹" + grossSalary);
    }

    public static void main(String[] args) {
        Employee e = new Employee();
        e.readDetails();
        e.calculateSalary();
        e.display();
    }
}
```

---

3. Student Fee Management System

```
import java.util.Scanner;

class Student {
    int studentId;
```

```
String studentName;
String course;
double feeBalance = 50000;

void readDetails() {
  Scanner sc = new Scanner(System.in);

  System.out.print("Enter Student ID: ");
  studentId = sc.nextInt();
  sc.nextLine();

  System.out.print("Enter Student Name: ");
  studentName = sc.nextLine();

  System.out.print("Enter Course: ");
  course = sc.nextLine();
}

void payFee() {
  Scanner sc = new Scanner(System.in);

  System.out.print("Enter Fee Amount to Pay: ");
  double amount = sc.nextDouble();

  feeBalance -= amount;
}

void checkBalance() {
  System.out.println("Remaining Fee Balance: ₹" + feeBalance);
}

void display() {
  System.out.println("\n----- Student Details -----");
  System.out.println("Student ID: " + studentId);
  System.out.println("Student Name: " + studentName);
  System.out.println("Course: " + course);
  System.out.println("Fee Balance: ₹" + feeBalance);
}
```

```
    public static void main(String[] args) {
        Student s = new Student();
        s.readDetails();
        s.payFee();
        s.checkBalance();
        s.display();
    }
}
```

---

4. Electricity Bill Management

```
import java.util.Scanner;

class ElectricityBill {
    int consumerNumber;
    String consumerName;
    String connectionType;
    int unitsConsumed;

    double billAmount;

    void readDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Consumer Number: ");
        consumerNumber = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Consumer Name: ");
        consumerName = sc.nextLine();

        System.out.print("Enter Connection Type (Domestic/Commercial): ");
        connectionType = sc.nextLine();

        System.out.print("Enter Units Consumed: ");
        unitsConsumed = sc.nextInt();
```

```
    }

    void calculateBill() {
        if (connectionType.equalsIgnoreCase("Domestic"))
            billAmount = unitsConsumed * 5;
        else if (connectionType.equalsIgnoreCase("Commercial"))
            billAmount = unitsConsumed * 8;
        else
            System.out.println("Invalid Connection Type");
    }

    void display() {
        System.out.println("\n----- Electricity Bill -----");
        System.out.println("Consumer Number: " + consumerNumber);
        System.out.println("Consumer Name: " + consumerName);
        System.out.println("Connection Type: " + connectionType);
        System.out.println("Bill Amount: ₹" + billAmount);
    }

    public static void main(String[] args) {
        ElectricityBill e = new ElectricityBill();
        e.readDetails();
        e.calculateBill();
        e.display();
    }
}
```

---

5. Mobile Recharge System

```
import java.util.Scanner;

class MobileRecharge {
    String mobileNumber;
    String subscriberName;
    String serviceProvider;
    double accountBalance = 500;
```

```
void readDetails() {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter Mobile Number: ");
    mobileNumber = sc.nextLine();

    System.out.print("Enter Subscriber Name: ");
    subscriberName = sc.nextLine();

    System.out.print("Enter Service Provider: ");
    serviceProvider = sc.nextLine();
}

void recharge() {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter Recharge Amount: ");
    double amount = sc.nextDouble();

    accountBalance += amount;
}

void deductBalance() {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter Usage Amount: ");
    double usage = sc.nextDouble();

    if (usage <= accountBalance)
        accountBalance -= usage;
    else
        System.out.println("Insufficient Balance");
}

void display() {
    System.out.println("\n----- Subscriber Details -----");
    System.out.println("Mobile Number: " + mobileNumber);
    System.out.println("Subscriber Name: " + subscriberName);
```

```java
        System.out.println("Service Provider: " + serviceProvider);
        System.out.println("Updated Balance: ₹" + accountBalance);
    }

    public static void main(String[] args) {
        MobileRecharge m = new MobileRecharge();
        m.readDetails();
        m.recharge();
        m.deductBalance();
        m.display();
    }
}
```